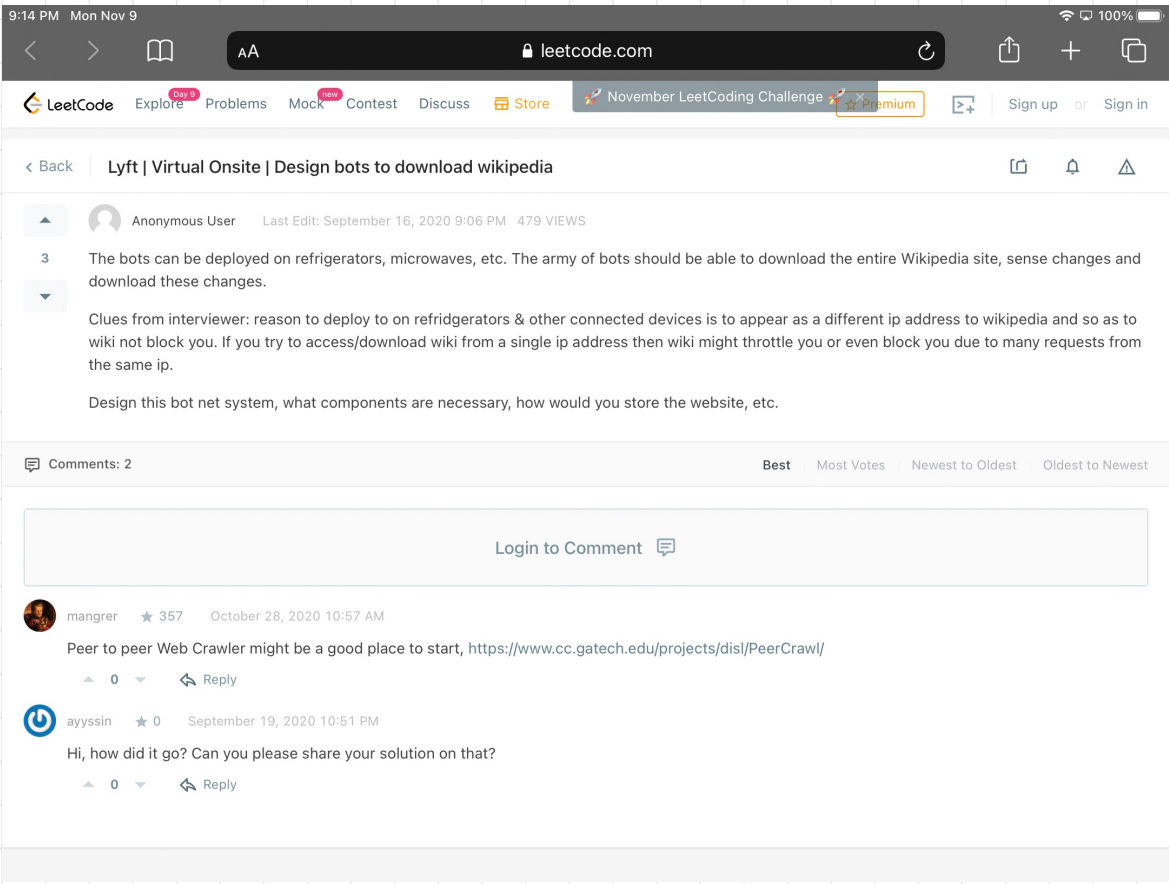


Wikipedia Crawler

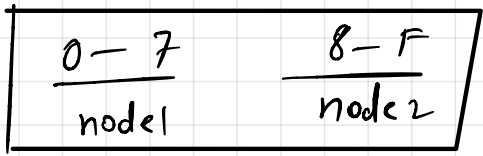
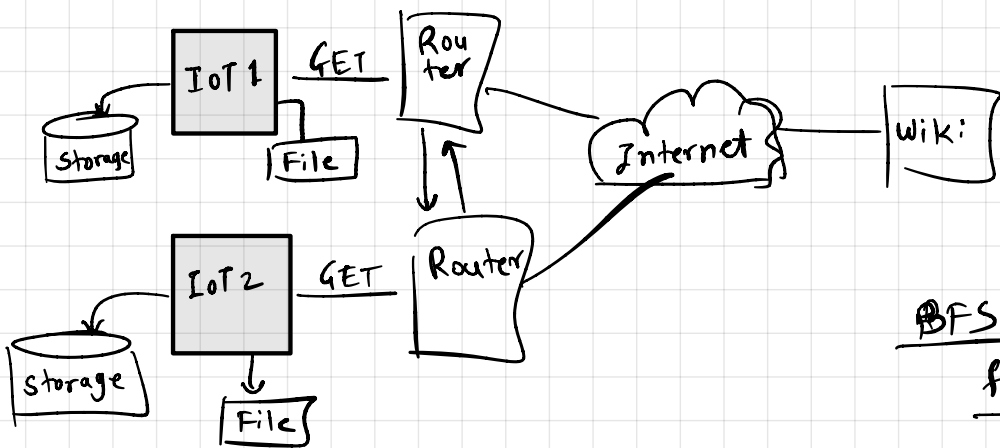
REQUIREMENTS

- Bot deployed on iot device
- Entire wikipedia
- Distributed crawling
- inform each other
- store content on each bot
- should not download duplicate
 - Accuracy?
- serverless chat system
- Distributed hash table
 - Each pod will own part of the hash table.
- How dynamo works (leaderless, multi-master db)
- Can't rely on one-bot
- multiple copies
- Broadcast == centralized?
- Registry server
 - Register a new bot
- P-to-P network
- Basic pub-sub system
- Gossip protocol
- Communication betⁿ 2-bots
- VPN?
- tit-for-tat protocol
- Bit-coin protocol
- chord protocol
- Download to Box :)
 - 10-copies
 - metadata of copy
- Bloom filter on all the bots



wikipedia.com/f.jpg

- 500 - million
- 20 - bytes



1000 devices

(CH) → (Range based distribution)

BFS

file ← queue

- 5 - mins

URL1 → hash1 ← 0 to 7 (IoT1)
URL2 → hash2 ← 8 to F (IoT2)
URL3 → hash3
⋮